Anshul Noori

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Education

Phillips Exeter Academy Exeter, NH

Expected June 2027

- **Computer Science:** CSC590 Game Playing AI (A), CSC590 Game Development (A)
- **Advanced Coursework:** MAT420–510 Calculus, CHE411–431 Accelerated Chemistry, PHY400 Modern Physics, REL460 Ethics of the Marketplace (A)
- **Senior Year:** INT455 Principles of Engineering and Design, MAT520 Calculus, PHY470 Introductory Astronomy, HIS410 U.S. History
- **AP Exams:** Calculus AB (5), Computer Science A (5), Physics 1 (5)

Stanford Online High School Online

Grades 6–8

- **Coursework:** OCS15 AP Computer Science A (A), OCS25 Data Structures and Algorithms in Java (A), OCS35 Advanced Topics in Computer Science: Computer Systems (A−/A)
- Data Structures and Computer Systems are taught at a post-AP, early-college level

Technical Skills

Languages: C, C++, Zig, Rust, Go, Python, Nix, OCaml, Kotlin, TypeScript, JavaScript, Java, C#, Swift, Objective-C, Lua, PHP, SQL, Assembly, Starlark, Bash, Zsh, PowerShell
GPU & Graphics: Vulkan, Metal, CUDA, Slang, SPIR-V, GLSL
Frameworks & Libraries: React, Astro, Node.js, Bun, Turborepo, FastAPI, Kotlin Multiplatform, Kotlin Coroutines, Tokio, gRPC, Protocol Buffers, Textual
Cloud & Infrastructure: Cloudflare, AWS, GCP, Azure, Stripe, E2B, Kubernetes, Docker, OpenTofu, GitLab, GitHub Actions, Buildkite, Buck2, Gradle, Linux, Git
Hardware: KiCad, STM32, QMK, PCB design

Experience

Chief Technology Officer December 2025 – Present

Topit AI

- Lead technology at an AI startup that helps people choose their best photos, overseeing system design, security, privacy, and image analysis
- Moved the company's code from GitHub to a self-hosted GitLab on AWS, with Google sign-in and automatic backups, and defined all cloud infrastructure as code
- Moved builds onto remote cloud machines with a shared cache, so a full build of the codebase takes 9 seconds when cached and 52 seconds from scratch
- Audited cloud spending and cut waste: each automated deploy now downloads about 16 KB instead of 230 MB of code, and tests skip parts of the code that did not change

Technical Consultant April 2026 – Present

PuroPlate

Open Source Contributions

Buck2, Meta's build system | *Rust* July 2026 – Present

- Let Go projects save build results to a shared cache so later builds can reuse them; in testing, 501 of 526 build steps were reused instead of rerun (#1418, merged)
- Building a feature (3,000+ lines) that sends live build and test results to the dashboards Bazel users already use, showing which parts of a build failed (upcoming)

niks3, Nix build cache server | *Go* October 2026 – Present

- Contributing changes, first built for Topit's cache, that let several caches share one storage bucket and let niks3 run on Cloudflare's servers (upcoming)

SPIRV-Cross, Khronos Group's shader translator | *C++* August 2026

- Added support for a 3D graphics setting that Apple's Metal lacks, so programs written for Vulkan render depth correctly on Apple devices; passed all 319 existing tests (#2666, merged)

MoltenVK, Khronos Group's Vulkan for Apple devices | *Objective-C++* April 2026

- Added a Vulkan graphics feature that controls depth clipping, so games and apps that need it can run on Macs and iPhones (#2712, merged)

NixOS on Apple Silicon, nix-community | *Nix* March 2026

- Added support for the Limine boot loader to the project that runs NixOS Linux on Apple Silicon Macs (#429, merged)

Projects

Silt, 3D game engine | *Zig, Vulkan* April 2026 – Present

- Building a game engine from scratch for Windows, macOS, and Linux; when Macs lacked a graphics feature it needed, I added that feature to MoltenVK
- Draws only as much detail as the screen can show, similar to Unreal Engine's Nanite, so large worlds stay fast
- Loads parts of the world in the background by predicting where the player is heading

Trading, automated trading system | *Rust, Python, SQL* July 2026 – Present

- Building a system that researches, tests, and runs stock and options trading strategies, with statistical checks against lucky results and human approval required before any real money is used

Homelab, home server cluster | *Kubernetes, NixOS* October 2026 – Present

- Turned a laptop into a home server running Kubernetes, set up entirely from config files in Git; it hosts my email system, automations, and AI models on its graphics card
- Set up private remote access with single sign-on, a password-manager-backed secret store, and isolated virtual machines that start in about 2 seconds for running untrusted code

IO80, mechanical keyboard | *Circuit Design, KiCad, Python* October 2026

- Designed an 88-key keyboard circuit board with per-key lighting, a volume knob, and a built-in fingerprint reader, reporting key presses 8,000 times per second
- Generated the board with Python scripts and checked it with circuit and signal simulations before manufacturing: every key works and none interfere in 5,000 random key combinations

Prometheus, custom keyboard | *Circuit Design, KiCad* December 2025 – Present

- Designed a keyboard and its circuit board from scratch, teaching myself electrical engineering along the way
- Combines mechanical switches with magnetic sensors that let users set how far each key must be pressed; grew from a 60% to a 75% layout over several versions

20/20 Vision, Minecraft mod | *Rust, Kotlin, Vulkan* April 2026 – Present

- Building a mod that lets players see much farther by drawing distant terrain at lower detail and moving the drawing work to the graphics card

Gum, robotics framework | *Kotlin* July 2026 – Present

- Replaced my FTC team's standard tools with an open-source framework, published for other teams, that makes robot code simpler to write and lets teams update code on the robot between matches

Whoof, Whoop for guide dogs | *Python, PyTorch, Hardware* July 2025 – Present

- Designing an AI model that predicts which guide-dog puppies will finish training, so nonprofits can focus limited resources on likely graduates
- Prototyping a sensor collar that tracks a puppy's movement, heart rate, breathing, and sounds, to combine with video

Alzheimer's Research Agent | *TypeScript, Python* October 2026 – Present

- Designing an open AI research assistant that reads Alzheimer's studies and datasets, links findings across papers, and is judged on results that matter for drug development

Hermes, medication safety app | *Python, TypeScript, React* January 2026

- Built solo in 24 hours at the Cline × Cerebras.AI hackathon (honorable mention against teams); warns users about dangerous drug combinations using FDA data on reported side effects

Pathways | *Kotlin*: schedule app for Exeter students that works offline on iOS, Android, and desktop February 2024

Game AI Pathfinding | *Rust, C++*: fast route-finding for game characters December 2025

14 kB Websites | *TypeScript, Astro*: pages small enough to load in one network round trip May 2026

ani-tui | *Python*: terminal app for browsing and tracking anime on AniList June 2026 – July 2026

NixOS Setup | *Nix*: my computers and a game server, rebuilt from config files February 2026 – Present

Y86 Assembler | *C#*: turns Y86 assembly language into machine code September 2023

Leadership & Service

Gentle Generations | *Co-Founder & President* 2020 – Present

- Co-founded a nonprofit that connects young people with senior citizens through service
- Let volunteers across the country and abroad design service projects around their own skills, including a month-long bake sale that raised over $1,000 for the Alzheimer's Association

FTC Team APEX 24513 | *Lead Programmer* September 2025 – Present

- Wrote the robot's software in Kotlin and Java for the 2025–26 season, covering driving, ball intake, sorting, and shooting
- Built autonomous routines that follow planned paths and use a camera to find targets, with battery-voltage compensation so the robot behaves the same as its battery drains

Guide Dog Puppy Raiser | *501(c)(3) guide dog nonprofit* 2025 – Present

- Raising and training Remington to become a guide dog for a person who is visually impaired

Alzheimer's Association | *Youth Board Chair* 2026 – Present

Nurturing Neurons | *Co-Founder* **Fit4Grit** | *CTO* **Birthday Dreams** | *Business Intern*

Awards

Computer Science Upper Classmen Award, Phillips Exeter
1517 Medici Project Grant, 1517 Fund
Honorable Mention, Cline × Cerebras.AI Hackathon
College Board National Recognition Program
National Merit Scholarship Program Qualifier
First-Year English Award, Phillips Exeter